Kickstart Your Zephyr-RTOS Embedded Journey

Next-Level Development mit Nordic und Future

March 2026

Nordic Semiconductor

Christian Kurz

Thomas Page

Angela Haug

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Agenda des Workshops

- Einleitung von Nordic zum *nRF Connect SDK*
- Hands-on Kapitel 1: IDE Einführung & Blinky Applikation
- Hands-on Kapitel 2: Bluetooth Peripheral LED & Button Service (LBS)
- Hands-on Kapitel 3: Nordic PMIC & Fuel Gauge Integration

We are Nordic

Simplifying lives through all things connected

Fabless semiconductor company

Founded in Norway, 1983

~1,350 employees

$511 MUSD revenue, CY 2024
OSEBX: NOD

ISO 9001 certified

Global presence

Complete solution

Faster time-to-market

Next-gen hardware	Embedded software	World-class support	Customer device	Cloud lifecycle services
ICs, SoCs, SiPs, PMICs	**nRF Connect SDK** Unified software	DevZone Developer community	Consumer	nRF CLOUD powered by Memfault
3rd party modules	Mobile Apps	DevAcademy Online hands-on trainings	Healthcare	Device management
Embedded SW stacks	Extensive SW/HW development tools	NORDIC SOLUTION PARTNER, NORDIC DESIGN PARTNER, NORDIC MODULE PARTNER Extensive technology partner program	Industrial	Embedded observability
				Location services

Bluetooth matter THREAD LTE-M NB-IoT nrf WiFi

Nordic product overview

Cloud support across all our wireless connectivity solutions

Short-Range	Cellular IoT	Wi-Fi 6 IoT	Power Management
nRF 54 SERIES Bluetooth, THREAD, matter, zigbee nRF CLOUD powered by Memfault	nRF 91 SERIES LTE-M, NB-IoT, GNSS nRF CLOUD powered by Memfault	nRF 70 SERIES Wi Fi, matter nRF CLOUD powered by Memfault	nPM FAMILY

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nRF Connect SDK

Introduction

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nRF Connect Platform Overview

nRF Connect

nRF Connect SDK
- Samples and Applications
- Middleware
- RTOS
- Libraries
- Hardware Drivers

nRF Connect for Desktop
- Quickstart
- Bluetooth Low Energy
- Programmer
- Cellular Monitor
- Power Profiler
- nPM PowerUP

nRF Connect for VS Code
- nRF Terminal
- nRF Debug
- Memory Report
- Command Line and GUI
- Create New Board Wizard

Mobile Applications
- nRF Connect for Mobile
- nRF Mesh
- nRF Toolbox
- nRF Blinky
- Device Manager

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nRF Connect SDK

Unified code base and toolchain for all Nordic SoCs

- Modern tools and SDK
- In the market since 2018
- Based on Zephyr RTOS
- Strong focus on open-source
- Future-proof, modular approach

- Single SDK for all Nordic product series
- Bluetooth v6 qualified Host and Controller stack since v2.8.0

Bluetooth THREAD matter ANT

zigbee Wi Fi LTE-M NB-IoT GPS

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Why the Zephyr RTOS?

- Zephyr is designed & built for low power wireless
- Zephyr is an independent technology project
  - Governed under the Linux Foundation
  - Contributions made by over 1000 industry embedded experts
- Zephyr is scalable
  - Very small configurations for memory constrained devices
  - Powerful, feature-rich, configurations for large memory, high-processing power devices (multiple MBs)
  - Designed for low power applications
- Zephyr includes a lot of fundamental pieces of code
  - Nordic can focus on other important features
- Nordic has been a member & contributor since 2016

Zephyr™ Project

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nRF Connect SDK Code Base

- Code base consists of several integrated repositories
  - Code coming from Nordic
  - Code coming from Open Source (OS) repositories

- Contains an RTOS, application code, connectivity protocols, wireless stacks, peripheral drivers & more

290+ Sensors Already Integrated

adt7420
adxl345
adxl362
adxl372
ak8975
amg88xx
ams_as5600
ams_iAQcore
apds9960
bma280
bmc150_magn
bme280
bme680
bmg160
bmi160
bmi270
bmm150
bmp388
bq274xx
ccs811

dht
dps310
grove
grow_r502a
hmc5883l
iis2dh
iis2dlpc

iis2iclx
iis2mdc
ite_vcmp_it8xxx2
lis2dh
lis2ds12
lm75
lps25hb
lsm303dlhc_magn

lsm6ds0
lsm6dsl
max17262
max30101
max31875
max44009
max6675
mchp_tach_xec
mpr
mpu9250
ms5607
ms5837

nrf5
nuvoton_adc_cmp_npcx
nxp_kin
opt300
pms7003
qdec_nrfx
qdec_sam
qdec_stm32
rpi_pico_temp
sgp40
sht3xd
sht4x
shtcx
si7006
si7055
si7060

si7210
sm351lt
stm32_temp
stm32_vbat
stmemsc
stts751
sx9500
th02
ti_hdc
ti_hdc20xx
tmp007
tmp108
tmp112
tmp116
vcnl4040
vl53l0x
wsen_hids
wsen_itds

github.com/zephyrproject-rtos/zephyr/tree/main/drivers/sensor

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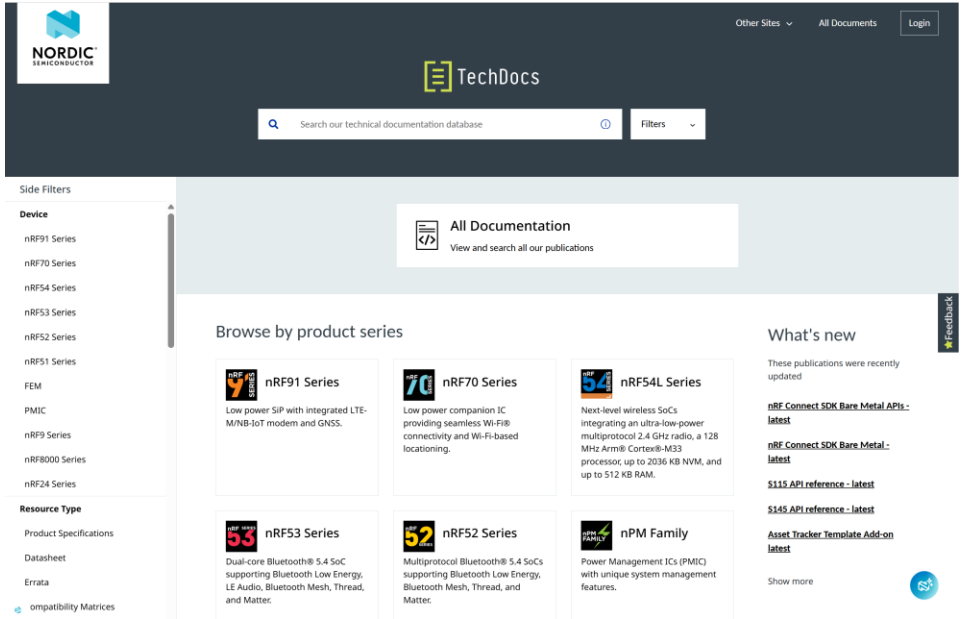
https://www.zephyrproject.org/wp-content/uploads/2026/01/Zephyr-Overview-20260107.pdf

Documentations & Help

Guidance

Where do I find documentation?

- Nordic created a new, unified website for its documentation called TechDocs:
  - www.docs.nordicsemi.com



- HW: Tiles represent the product series on the main page
- SW: Scroll down to “Quick Links”
  - Links to various repositories that are part of the nRF Connect SDK
  - (Entry Page: “nRF Connect SDK – latest”)

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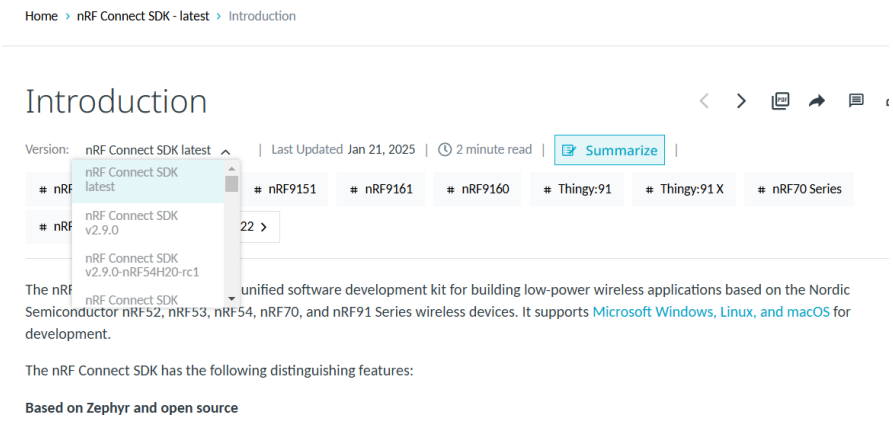
nRF Connect SDK Documentation

- nRF Connect SDK: [Documentation link]
- Ensure to view the documentation corresponding to your NCS version in use
  - Click on the version field / drop down menu
  - "Latest" refers to the current development branch (main branch)
  - Other versions listed as per SDK releases

- Note that not all sub pages list all SDK release versions, as the respective sub page might not exist in that SDK release



Have you seen this button?

It‘s found on following web pages:

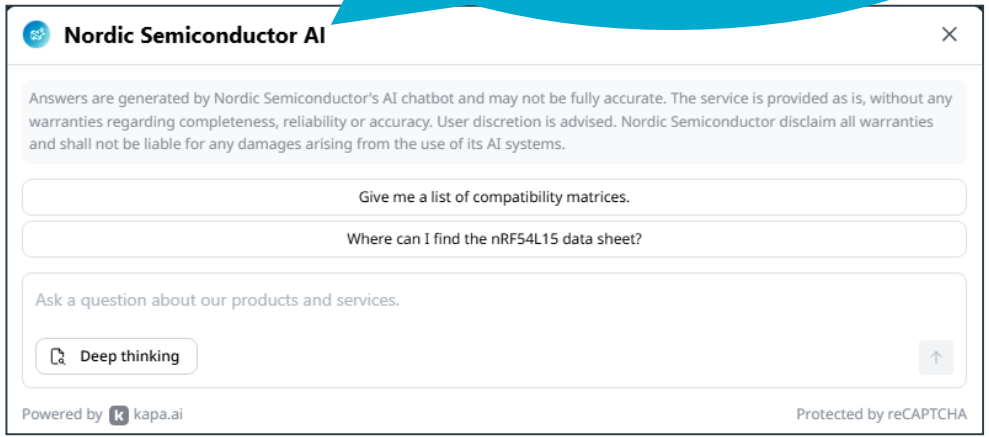
- Nordic Semiconductor‘s TechDocs
- Nordic Semiconductor‘s DevZone

Nordic Semiconductor AI Agent

- Answering questions about Nordic Semiconductor's products, solutions, and documentation.

I am specialized in answering questions about Nordic Semiconductor's products, solutions, and documentation. If you have questions about Nordic products or technologies, I am happy to help!

- AI Agent uses Nordic Semiconductor resources to look for an answer:
  - Official Nordic Semiconductor documentation
  - Nordic Semiconductor Blog posts
  - Nordic Semiconductor Forum Discussions
  - Nordic Semiconductor Support Resources



- Different Languages are supported

nRF Connect SDK

Deep Dive

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nRF Connect SDK Repositories

- nRF: Application & connectivity protocols
- nrfxlib: compiled libraries where Nordic cannot distribute source code
- nrfx: peripheral drivers
- Zephyr: RTOS & board configuration (OS)
- MCUboot: Secure Bootloader (OS)
- Other repositories
  - Trusted Firmware-M
  - Matter
  - etc.

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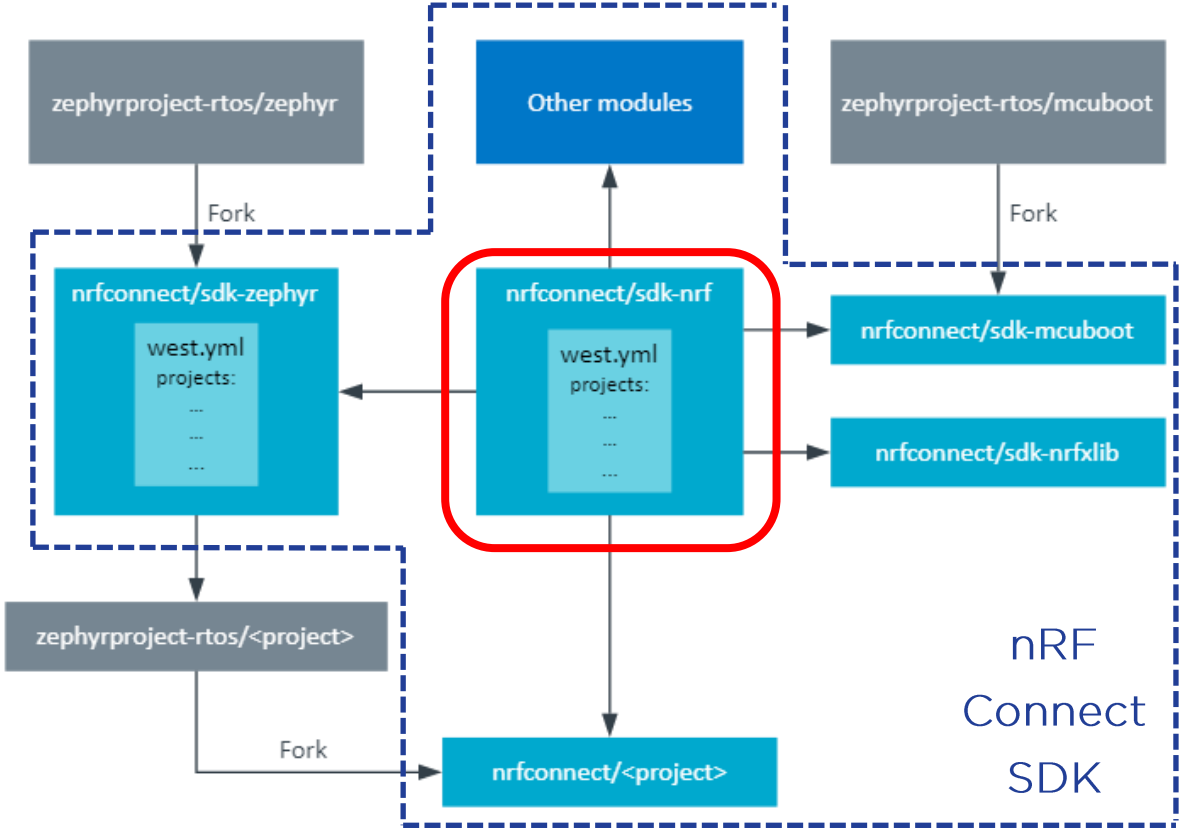
nRF Connect SDK Structure

- *nRF Connect SDK* is structured as star topology
  → sdk-nrf repo is the center
  (github.com/nrfconnect/sdk-nrf)

- sdk-nrf repo contains manifest file west.yml
  → Lists and points to all other repos included in the SDK



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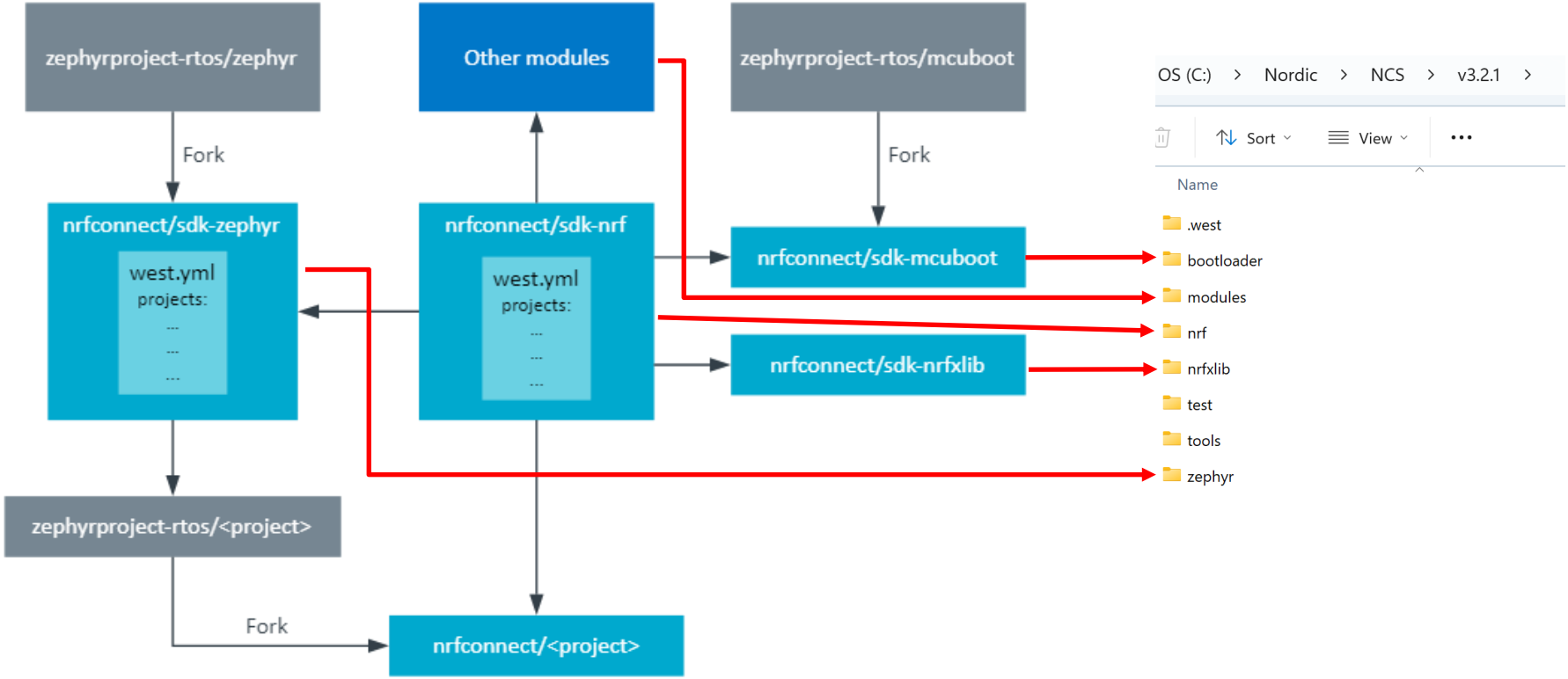
Repository Structure vs. Installation Paths



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Code Examples

Code Examples can be found from Nordic or Zephyr.

This PC > OS (C:) > Nordic > NCS > v3.2.3 > nrf > samples >

Sort View ...

Name	Date modified	Type	Size
app_event_manager	3/2/2026 12:59 PM	File folder	
app_event_manager_profiler_tracer	3/2/2026 12:59 PM	File folder	
app_jwt	3/2/2026 12:59 PM	File folder	
basic	3/2/2026 12:59 PM	File folder	
benchmarks	3/2/2026 12:59 PM	File folder	
bluetooth	3/2/2026 12:59 PM	File folder	
bootloader	3/2/2026 12:59 PM	File folder	
caf	3/2/2026 12:59 PM	File folder	
caf_sensor_manager	3/2/2026 12:59 PM	File folder	
cellular	3/2/2026 12:59 PM	File folder	
common	3/2/2026 12:59 PM	File folder	
crypto	3/2/2026 12:59 PM	File folder	
debug	3/2/2026 12:59 PM	File folder	
dect	3/2/2026 12:59 PM	File folder	
dfu	3/2/2026 12:59 PM	File folder	
edge_impulse	3/2/2026 12:59 PM	File folder	

Nordic examples → **nrf/samples**

This PC > OS (C:) > Nordic > NCS > v3.2.3 > zephyr > samples >

Sort View ...

Name	Date modified	Type	Size
application_development	3/2/2026 12:59 PM	File folder	
arch	3/2/2026 12:59 PM	File folder	
basic	3/2/2026 12:59 PM	File folder	
bluetooth	3/2/2026 12:59 PM	File folder	
boards	3/2/2026 12:59 PM	File folder	
cpp	3/2/2026 12:59 PM	File folder	
data_structures	3/2/2026 12:59 PM	File folder	
drivers	3/2/2026 12:59 PM	File folder	
hello_world	3/2/2026 12:59 PM	File folder	
kernel	3/2/2026 12:59 PM	File folder	
modules	3/2/2026 12:59 PM	File folder	
net	3/2/2026 12:59 PM	File folder	
philosophers	3/2/2026 12:59 PM	File folder	
posix	3/2/2026 12:59 PM	File folder	
psa	3/2/2026 12:59 PM	File folder	
sensor	3/2/2026 12:59 PM	File folder	
shields	3/2/2026 12:59 PM	File folder	

Zephyr examples → **zephyr/samples**

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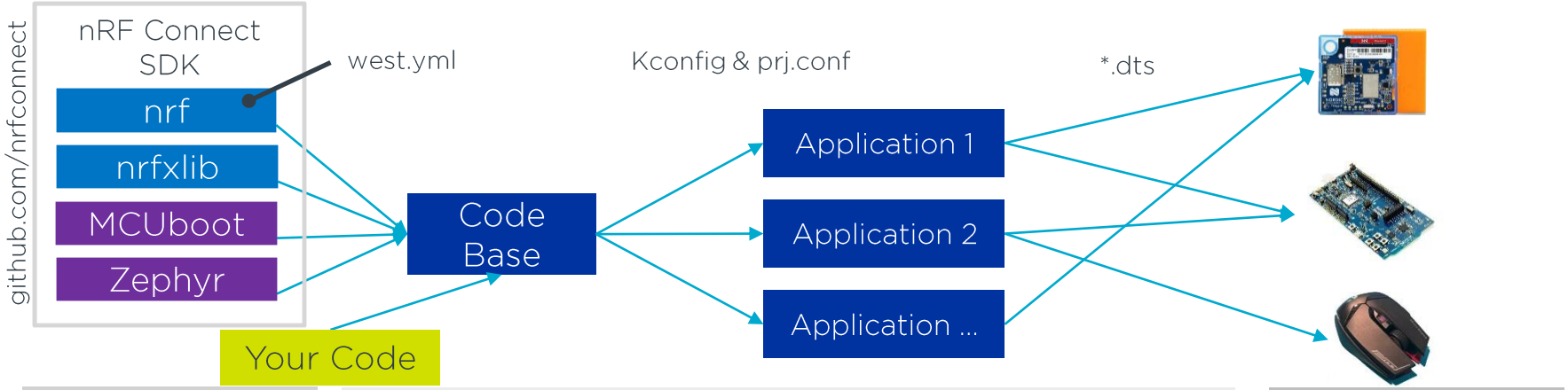
Manage Source Code and Configurations

West
Multi-repository management tool

Kconfig
Source module / feature configuration for compile

Device Tree
Target Board / Device description

Clone / update → Configure features → Configure target



Zephyr Application Types

Repository

An app located in Zephyr repository is referred to as a *Zephyr Repository Application* (e.g. hello_world sample)

```
zephyrproject/
├── .west/
│   └── config
└── zephyr/
    ├── arch/
    ├── boards/
    ├── cmake/
    ├── samples/
    │   ├── hello_world/
    │   └── ...
    ├── tests/
    └── ...
```

West Workspace

An app located within a workspace but outside the Zephyr repository is referred to as a *Zephyr Workspace Application*

```
zephyrproject/
├── .west/
│   └── config
├── zephyr/
├── bootloader/
├── modules/
├── tools/
├── <vendor/private-repos>/
└── applications/
    └── app/
```

Freestanding

Zephyr Freestanding Application is an app located outside a Zephyr workspace (e.g. nRF Connect SDK installation)

```
<home>/
├── zephyrproject/
│   ├── .west/
│   │   └── config
│   ├── zephyr/
│   ├── bootloader/
│   ├── modules/
│   └── ...
│
└── app/
    ├── CMakeLists.txt
    ├── prj.conf
    └── src/
        └── main.c
```

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Zephyr/NCS Application (minimal project)

```
Nordic
├── Workspace
│   ├── MyProject
│   │   ├── CMakeLists.txt
│   │   ├── prj.conf
│   │   └── src
│   │       └── main.c
│   │
```

CMAKE

File: CMakeLists.txt

```
cmake_minimum_required(VERSION 3.20.0)
find_package(Zephyr REQUIRED HINTS $ENV{ZEPHYR_BASE})
project(hello_world)
target_sources(app PRIVATE src/main.c)
```

CKONFIG

File: prj.conf

```
# nothing here! (default settings, e.g. CONFIG_GPIO=y)
```

Source Files

Application (user code) + selected software modules from Code Base (SDK)

File: src/main.c

```
#include <zephyr/kernel.h>

int main(void){
    printk("Hello World!\n");
    return 0;
}
```

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Zephyr/NCS Application – optional parts

```
Nordic
├── Workspace
│   ├── MyProject
│   │   ├── CMakeLists.txt
│   │   ├── prj.conf
│   │   ├── Kconfig
│   │   ├── sysbuild.conf
│   │   ├── VERSION
│   │   ├── board
│   │   ├── snippets
│   │   ├── sysbuild
│   │   └── src
│   │       └── main.c
│   │
├── MyProject_RISCV
│   ├── CMakeLists.txt
│   …
│
├── ncs
│   ├── bootloader
│   │   ├── mcuboot
```

- **Kconfig**
  Configuration System. Allows user defined Kconfig symbols for custom application (used in *.conf files).
- **VERSION**
  Application Version Management. Version is accessible from Source Code, Kconfig, Cmake, and Shell.
- **Board**
  Overlay DTS file allows to change pre-defined settings of a board. Or an own board can be defined from scratch.
- **Snippets**
  save common configuration separately when it applies to multiple different applications.
- **Sysbuild.conf + sysbuild folder**
  used to combine multiple project builds together, e.g. build application together with MCUboot.